

J: Just Palindromes!

Source file name: just.c, just.cpp, just.java, or just.py

Author: Rodrigo Cardoso

Ana is fascinated by the idea of finding *palindromes*, i.e., sentences that read the same backward and forward once you ignore spaces, punctuation, letter case, and letter characters outside the English alphabet. For example, the following four sentences are palindromic:

Go deliver a dare vile dog.

Cigar? Toss it in a can. It is so tragic.

Never odd or even!

A man, a plan, a canal: Panama!

Note that in a palindrome, only letter characters from the English alphabet are allowed, case does not matter, and all non-letter characters (punctuation, spaces, apostrophes, etc.) are ignored.

On the other hand, the following sentences are not palindromes, even if case and non-letter characters are ignored:

This is a palindrome.

Hello, world!

Help Ana by writing a program that reads a sentence and reports whether it is a palindrome.

Input

The input consists of multiple test cases. Each test case is a single line containing a sentence S of length N ($0 \leq N < 1\,000$) and made only of printable ASCII characters (i.e., characters with ASCII code between 32 and 126, inclusive). The input ends with a line containing a single asterisk '*', which must not be processed.

The input must be read from standard input.

Output

For each test case, output a single line with the letter character 'Y' if S is a palindrome and 'N' if not.

The output must be written to standard output.

Sample Input	Sample Output
Go deliver a dare vile dog.	Y
Cigar? Toss it in a can. It is so tragic.	Y
A man, a plan, a canal: Panama!	Y
This is a palindrome.	N
.----.-	Y
*	